Research Topics Related to Pancreas Segmentation

Sugimoto Lab.

① Signal-Based Interpretation of Segmentation Evaluation Metrics and Uncertainty-Aware Assessment of Segmentation Reliability

This research aims to advance the evaluation of medical image segmentation by shifting the focus from accuracy comparison to interpretability and quality assessment of segmentation results. It clarifies how different error structures are reflected -or overlooked - by standard evaluation metrics.

First, segmentation errors are decomposed into interpretable components such as boundary deviation, localized defects, small-region omission, and global volume bias. Standard metrics (e.g., Dice, IoU, and HD95) are then reinterpreted along clinically meaningful dimensions, such as boundary error corresponding to clinical risk and volume bias corresponding to decision bias. This signalization framework demonstrates that similar metric values may arise from fundamentally different geometric and clinical failure modes.

Second, we introduce an uncertainty-aware quality assessment framework that integrates spatial uncertainty patterns with metric behavior to derive a quality score representing segmentation reliability. This score is used to automatically estimate whether expert review or manual correction is likely required, enabling physician-in-the-loop quality control without replacing clinical judgment.

② Controlled Error Analysis of Segmentation Metrics Under Clinically Relevant Failure Modes

This research evaluates segmentation metrics through a controlled error injection framework. Starting from ground-truth segmentation masks. Predefined geometric error patterns, including boundary shifts, localized defects, complete omission of small tumors, and global volume over- or under-estimation, are systematically generated.

By applying standard evaluation metrics (Dice, IoU, and HD95) to these synthetically perturbed masks, how each metric responds to specific error types is analyzed. This approach allows us to characterize the sensitivity and blind spots of commonly used metrics, revealing cases where similar scores correspond to substantially different geometric or clinically relevant errors. The framework decouples metric behavior from model-specific biases and provides an interpretable basis for understanding how segmentation errors translate into metric values, with direct implications for clinical reliability and downstream decision-making.

③ Pancreatic Tumor Segmentation with Low Estimation Accuracy and Limited Annotations

This research aims to address pancreatic tumor segmentation, a task characterized by persistently low estimation accuracy and severe annotation scarcity, by establishing learning strategies that effectively leverage limited labeled data and large amounts of unlabeled data. Pancreatic tumors are typically very small, exhibit high variability in shape and location, and show low contrast with surrounding tissues on CT images, making reliable segmentation difficult even under fully supervised settings.

First, investigate semi-supervised and weakly supervised learning approaches in which a small number of pixel-level tumor annotations are combined with coarse or weak labels (e.g., scribbles, bounding boxes, presence/absence labels) and high-confidence pseudo-labels. These methods aim to progressively extract tumor-related information from unlabeled cases while reducing dependence on fully annotated tumor masks.

Second, explore self-supervised learning with masked pre-training using large-scale unlabeled abdominal CT data. Through reconstruction- and prediction-based pretext tasks, the model is encouraged to learn representations that capture pancreatic anatomy, surrounding structures, intensity distributions, and boundary characteristics. These representations are then transferred to downstream pancreatic tumor segmentation to improve robustness and stability under extremely limited annotation conditions.

Finally, from a few-shot segmentation perspective, study approaches that enable rapid adaptation to new pancreatic tumor types or rare cases using only a handful of annotated examples. By leveraging meta-learning or prototype-based representations, the methods aim to efficiently capture highly localized and heterogeneous tumor features.

Through this study, we seek to establish annotation-efficient segmentation strategies that improve the practical feasibility of pancreatic tumor segmentation and support future applications in diagnostic assistance and preoperative planning.